

Q1 Assume that the diameter of the cross-section of the wire in a paper clip is 1 mm

$$\text{Cross-sectional area of the wire in a paper clip} = \pi \left(\frac{0.001}{2} \right)^2 = 7.85 \times 10^{-7} \text{ m}^2 \approx 8 \times 10^{-7} \text{ m}^2$$

Answer: C